

unit 7

- **Evolution (Charles Darwin—"Descent with Modification")**—gradual change over time to better fit an environment. Individuals do not evolve, populations do.
- **Adaptation**—inherited trait that makes an organism more "fit" for its environment
- **"Fitness"**—the likelihood that an organism will survive and reproduce. "Fit" traits are favored by natural selection.
- **Niche**—specific role or function of an organism in its environment.
- **Habitat**—where an organism lives in its environment.
- **Natural Selection**—the environment favors traits in individuals who are more likely to survive and reproduce, and thus the trait is naturally selected for
- **Artificial Selection**—human breeding of organisms to produce desired traits
- **Overshooting**—populations will produce too much offspring than the environment can support, meaning that many new offspring will die. The ones who survive are naturally selected for.
- **Requirements for Natural Selection**—competition, adaptation, heritability, variation
 - **Competition**—more offspring produced than the environment can support
 - **Variation**—offspring produced with traits that differ from one another
 - **Adaptation**—some adaptations must be more "fit" than others
 - **Hereditary**—must be genetic and able to be inherited by the next generation
- **Homologies/Homologous Structures (NOT HOMOLOGUES)**—similarities between species because of common ancestry, and so they have morphological/anatomical similarities. Same anatomy, possibly a different function.
- **Vestigial Structures**—structures that are now "useless" but were once useful and had a purpose in a common ancestor
- **Analogous Structures**—structures with a similar function but a different anatomy
 - **Convergent Evolution**—through different manners of evolution (different mutations, different anatomy), different species evolved the same trait to suit an environment
- **Comparative embryology**—all eukaryotic embryos look very similar, homologies found in embryos not seen in adults (ex: tail bone in humans)
- **Fossil record**—provides evidence of species extinction, changes within a group over time (natural selection), origin of new groups
- **DNA Sequencing**—nucleotide sequences can be compared between organisms to determine common ancestry
- **Gene pool**—all the alleles for every gene for every individual in a population
- **Population**—localized group of individuals of one species
- **Genetic variation**—lots of differences among a population

*** See Hardy-Weinberg Equilibrium on the Math Sheet.

• **Hardy-Weinberg Equilibrium**—hypothetical population that is not evolving

1. No Mutations
2. Random Mating (No Sexual Selection)
3. No Gene Flow
4. Large Population
5. No Natural Selection

• **Microevolution**—change within the species level, slow, over time, small changes

• **Genetic Drift**—allele frequencies in a population change rapidly due to random change, not due to natural selection

- ▶ **Founder Effect**—a few individuals become separated from their population, so they found a new population. The new gene pool has shrunk.
- ▶ **Bottleneck Effect**—a natural disaster or other event wipes out lots of members of a population, both reducing the gene pool and genetic diversity.

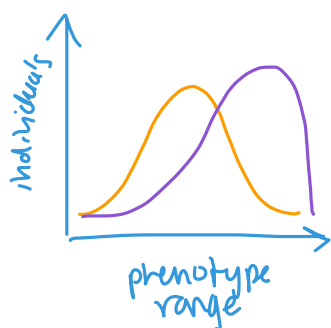
• **Gene Flow**—immigration and emigration between populations, ensures that speciation does not happen, can also change allele frequencies

• **Natural Selection**—acts on an organism's phenotype, not genotype.

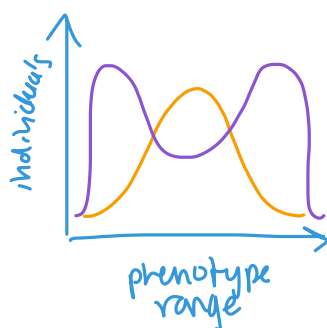
• **Heritable Variation**—germ-line mutations and sexual reproduction

• **Natural Selection Acts on a Population in One of Three Ways:**

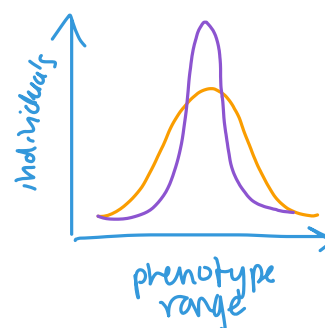
1. **Directional Selection**—natural selection favors one phenotype
2. **Disruptive Selection**—natural selection favors all but the intermediate
3. **Stabilizing Selection**—natural selection favors the intermediate phenotype



Directional



Disruptive



Stabilizing

• **Heterozygous Advantage**—natural selection will retain two alleles because the heterozygous phenotype is “fit”

• **Sexual Selection**—females (typically) select males as mates, so males will try to impress females

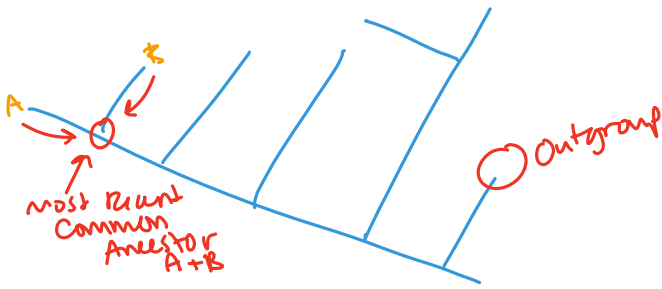
• **Sexual Dimorphism**—different phenotypes among males and females for the purpose of attracting a mate

• **Biological Species Concept**—species is a group of populations that can interbreed and produce viable, fertile offspring

• **Reproductive Isolation**—biological barriers impeding two organisms from producing viable, fertile offspring

- ▶ **Prezygotic barriers**—prevent viable, fertile offspring before fertilization
 - **Habitat Isolation**—two species do not encounter each other because they live in different habitats, and thus they cannot interact and breed
 - **Temporal Isolation**—two species mate in different seasons or at different times
 - **Behavioral Isolation**—behaviors which attract mates are different between species such as courtship rituals.
 - **Mechanical Isolation**—reproductive organs have morphological differences, preventing breeding
 - **Gametic Isolation**—the sperm cannot fertilize the egg
- ▶ **Postzygotic Barriers**—occur after fertilization
 - **Reduced hybrid fertility**—hybrid cannot have offspring (ex: mule)
 - **Reduced hybrid viability**—hybrid does not complete development or is frail and likely to die
 - **Hybrid breakdown**—hybrids can breed, but then hybrid offspring are weak or sterile
- **Morphological Species Concept**—species is defined by its morphological (structural) features
- **Ecological Species Concept**—species is defined by its ecological niche
- **Phylogenetic Species Concept**—a single branch on a phylogenetic tree, a single group of organisms that share a common ancestor
- **Speciation**—one species splits into two, driven by reproductive isolation
- **Recent Common Ancestor**—common ancestor shared by two species before speciation occurred
- **Macroevolution**—big changes on a species level over time
- **Allopatric Speciation**—population is geographically split into two populations.
- **Sympatric Speciation**—gene flow is reduced between groups for a reason other than geographic separation
 - ▶ **Polyploidy**—presence of extra chromosomes (no disjunction) that can create a new species
 - ▶ **Habitat Differentiation**—new ecological niches, so populations stop interacting because of their roles
 - ▶ **Sexual Selection**—certain females desire mates with particular phenotypes, leading to a split—speciation
- **Phylogenetic Tree/Cladogram**—branching tree diagram that shows evolutionary relationships between organisms
- **Homologies/Shared Derived Characteristics**—number of shared homologies determines the position of organisms on the cladogram
- **Node**—point on a phylogenetic tree that represents speciation
- **Outgroup**—species that shares the least homologies with every other species on the tree
- **Recent Common Ancestor**—determines how many homologies and how related two species are relative to the other organisms on the tree
- **Maximum Parsimony**—calculating the simplest (least amount of speciation, evolutionary derived characteristics) phylogenetic tree to fit modern

information



- **Anaerobic Atmosphere**—Earth's original atmosphere lacked oxygen but had an abundance of volcanic chemicals due to increased volcanic activity
- **Synthesis of Organic Molecules**—Miller-Urey experiment proved that a lightning strike could turn gases in Earth's anaerobic atmosphere into organic compounds
- **Macromolecule Construction**—how did organic compounds turn into complex macromolecules according to the evolution theory?
 - **Organic Soup Model**—accumulation of organic molecules in water eventually combined and led to macromolecules
 - **Clay Hypothesis**—when dripping monomers onto clay, they can spontaneously form polymers
- **Protocells**—early abiotic fluid-filled vesicles with a membrane that could internalize metabolic reactions, reproduce, and maintain an internal environment of certain chemicals
- **RNA World Hypothesis**—RNA can fold into ribozymes that can duplicate RNA, so RNA was probably the first genetic information.
- **Extinction**—inability of a species to survive in a changing environment
- **Mass Extinction Event**—lots of species go extinct worldwide in a short period of time
- **Geologic Record**—uses fossils to determine Earth's age and timeline, divided by mass extinction events.
- **Cyanobacteria**—first photosynthesizing autotrophs that used Earth's anaerobic atmosphere and CO₂ to produce food and O₂, slowing making Earth's atmosphere aerobic
- **Genetic Diversity**—larger populations with greater genetic diversity have a better chance at surviving rather than going extinct
- **Adaptive Radiation**—relatively quick speciation of many distinct species from a common ancestor